EnduranceLife API (0.1.0)

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RESTful API for managing endurance-sport training data parsed from .fit files (Coros, Garmin, etc.), daily nutrition/recovery metrics, and physiological trend tracking.

Authentication

Register a new user

Create a new user account.

- **username**: 3–50 characters, must be unique.
- **password**: 6–100 characters, stored as a bcrypt hash.

Returns the created user (without the password hash).

REQUEST BODY SCHEMA: application/json
required

`username` required	string (Username) [3 .. 50] characters
`password` required	string (Password) [6 .. 100] characters

Responses

> 201

Successful Response

— 409

Username already taken

> 422

Validation Error

POST /auth/register

Request samples

Payload

Content type
application/json

Copy

```
{
    "username": "david",
    "password": "strongpass123"
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "id": 0,
    "username": "string",
    "is_active": true
}
```

Login and get a JWT token

Authenticate with username and password.

Returns a Bearer JWT token valid for 30 minutes (configurable via the `TOKEN_EXPIRE_MINUTES` environment variable).

Use the token in the `Authorization: Bearer <token>` header for protected endpoints.

REQUEST BODY SCHEMA: application/x-www-form-urlencoded
required

`grant_type` >	Grant Type (string) or Grant Type (null) (Grant Type)
`username` required	string (Username)
`password` required	string (Password)
`scope`	string (Scope) Default: `""`
`client_id` >	Client Id (string) or Client Id (null) (Client Id)
`client_secret` >	Client Secret (string) or Client Secret (null) (Client Secret)

Responses

> 200

Successful Response

— 401

Invalid credentials

> 422

Validation Error

POST /auth/login

Response samples

200 422

Content type
application/json

Copy

```
{
    "access_token": "string",
    "token_type": "bearer",
    "username": "string"
}
```

Logout (invalidate token)

Invalidate the current JWT token by adding it to the blacklist.

After logout, the same token cannot be used for authenticated requests. A new token must be obtained via `/auth/login`.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200**

Successful Response

POST /auth/logout

Response samples

200

Content type
application/json

Copy

```
null
```

Get current user profile

Return the profile of the currently authenticated user.

Requires a valid Bearer token in the Authorization header.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> 200

Successful Response

GET /auth/me

Response samples

200

Content type
application/json

Copy

```
{
    "id": 0,
    "username": "string",
    "is_active": true
}
```

Activities

Create Activity

Import a new activity record via JSON body.

If the `source_file` already exists in the database, a 409 Conflict is returned to prevent duplicate imports.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

`pid` required	integer (Pid) User / person ID
`source_file` required	string (Source File) Original .fit filename (used as dedup key)
`date` required	string <date> (Date)
`start_time` required	string <date-time> (Start Time) Activity start time in UTC

start_lat >	Start Lat (number) or Start Lat (null) (Start Lat) Starting latitude
start_lng >	Start Lng (number) or Start Lng (null) (Start Lng) Starting longitude
distance_km required	number (Distance Km)
duration_min required	number (Duration Min) Total duration in minutes
type required	string (Type) Activity type: 'Run' or 'Ride'
temperature >	Temperature (number) or Temperature (null) (Temperature)
humidity >	Humidity (number) or Humidity (null) (Humidity)
air_pressure >	Air Pressure (number) or Air Pressure (null) (Air Pressure)
altitude_gain >	Altitude Gain (number) or Altitude Gain (null) (Altitude Gain)
avg_heart_rate >	Avg Heart Rate (integer) or Avg Heart Rate (null) (Avg Heart Rate)
max_heart_rate >	Max Heart Rate (integer) or Max Heart Rate (null) (Max Heart Rate)
avg_pace_sec >	Avg Pace Sec (integer) or Avg Pace Sec (null) (Avg Pace Sec) Avg pace in sec/km
avg_cadence >	Avg Cadence (integer) or Avg Cadence (null) (Avg Cadence)
training_load >	Training Load (number) or Training Load (null) (Training Load) Coros training load metric
hr_array_json >	Hr Array Json (string) or Hr Array Json (null) (Hr Array Json) JSON array string of HR time-series
pace_array_json >	Pace Array Json (string) or Pace Array Json (null) (Pace Array Json) JSON array string of pace time-series

Responses

> 201

Successful Response

> 422

Validation Error

POST /activities/

Request samples

Payload

Content type
application/json

Copy


```
{
    "pid": 0,
    "source_file": "string",
    "date": "2019-08-24",
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string"
```

```
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "pid": 0,
    "source_file": "string",
    "date": "2019-08-24",
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string",
    "id": 0
}
```

List Activities

Retrieve a paginated list of activities scoped to the current user.

Supports optional filtering by `type`, and a date range (`date_from`, `date_to`) for efficient front-end chart windows.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`type` >	Type (string) or Type (null) (Type) Filter by activity type (Run / Ride)
`date_from` >	Date From (string) or Date From (null) (Date From) Start of date range (inclusive)
`date_to` >	Date To (string) or Date To (null) (Date To) End of date range (inclusive)
`skip`	integer (Skip) `>= 0` Default: `0` Number of records to skip
`limit`	integer (Limit) `[ 1 .. 200 ]` Default: `50` Max records to return

Responses

> **200**

Successful Response

> **422**

Validation Error

```
GET /activities/
```

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
  - {
        "pid": 0,
        "source_file": "string",
        "date": "2019-08-24",
        "start_time": "2019-08-24T14:15:22Z",
        "start_lat": 0,
        "start_lng": 0,
        "distance_km": 0,
        "duration_min": 0,
        "type": "string",
        "temperature": 0,
        "humidity": 0,
        "air_pressure": 0,
        "altitude_gain": 0,
        "avg_heart_rate": 0,
        "max_heart_rate": 0,
        "avg_pace_sec": 0,
        "avg_cadence": 0,
        "training_load": 0,
        "hr_array_json": "string",
        "pace_array_json": "string",
        "id": 0
    }
]
```

Upload a .fit file and auto-parse it into an Activity record

Upload a `.fit` file directly through Swagger UI or any HTTP client.

The file is parsed in-memory using `fitdecode` and the raw bytes are stored in the database (`fit_file_blob`) for later re-parsing or download. Extracted fields include distance, duration, heart rate, pace, GPS coordinates, cadence, and time-series arrays.

- Returns **201** with the created Activity on success.

- Returns **409** if a file with the same name was already imported.
- Returns **422** if the file contains no workout session data.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: multipart/form-data
required

`file` required	string <binary> (File) A .fit file exported from Coros/Garmin

Responses

> **201**

Successful Response

> **422**

Validation Error

POST /activities/upload

Response samples

201 **422**

Content type
application/json

Copy


```
{
    "pid": 0,
    "source_file": "string",
    "date": "2019-08-24",
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
```

```
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string",
    "id": 0
}
```

Get Activity

Retrieve a single activity by its primary key, scoped to current user.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

activity_id required	integer (Activity Id)

Responses

> **200**

Successful Response

> 422

Validation Error

GET /activities/{activity_id}

Response samples

200 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "source_file": "string",
    "date": "2019-08-24",
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string",
    "id": 0
}
```

Update Activity

Update an existing activity. Only fields present in the request body (non-None) are applied, allowing partial updates. Scoped to current user.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

activity_id required	integer (Activity Id)

REQUEST BODY SCHEMA: application/json
required

pid >	Pid (integer) or Pid (null) (Pid)
source_file >	Source File (string) or Source File (null) (Source File)
date	null (Date)
start_time >	Start Time (string) or Start Time (null) (Start Time)
start_lat >	Start Lat (number) or Start Lat (null) (Start Lat)
start_lng >	Start Lng (number) or Start Lng (null) (Start Lng)
distance_km >	Distance Km (number) or Distance Km (null) (Distance Km)
duration_min >	Duration Min (number) or Duration Min (null) (Duration Min)
type >	Type (string) or Type (null) (Type)
temperature >	Temperature (number) or Temperature (null) (Temperature)
humidity >	Humidity (number) or Humidity (null) (Humidity)
air_pressure >	Air Pressure (number) or Air Pressure (null) (Air Pressure)
altitude_gain >	Altitude Gain (number) or Altitude Gain (null) (Altitude Gain)

avg_heart_rate >	Avg Heart Rate (integer) or Avg Heart Rate (null) (Avg Heart Rate)
max_heart_rate >	Max Heart Rate (integer) or Max Heart Rate (null) (Max Heart Rate)
avg_pace_sec >	Avg Pace Sec (integer) or Avg Pace Sec (null) (Avg Pace Sec)
avg_cadence >	Avg Cadence (integer) or Avg Cadence (null) (Avg Cadence)
training_load >	Training Load (number) or Training Load (null) (Training Load)
hr_array_json >	Hr Array Json (string) or Hr Array Json (null) (Hr Array Json)
pace_array_json >	Pace Array Json (string) or Pace Array Json (null) (Pace Array Json)

Responses

> 200

Successful Response

> 422

Validation Error

PUT /activities/{activity_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "source_file": "string",
    "date": null,
```

```
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string"
}
```


Response samples

200 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "source_file": "string",
    "date": "2019-08-24",
    "start_time": "2019-08-24T14:15:22Z",
    "start_lat": 0,
    "start_lng": 0,
    "distance_km": 0,
    "duration_min": 0,
    "type": "string",
    "temperature": 0,
    "humidity": 0,
    "air_pressure": 0,
    "altitude_gain": 0,
    "avg_heart_rate": 0,
```

```
    "max_heart_rate": 0,
    "avg_pace_sec": 0,
    "avg_cadence": 0,
    "training_load": 0,
    "hr_array_json": "string",
    "pace_array_json": "string",
    "id": 0
}
```

Delete Activity

Delete an activity by its primary key. Scoped to current user.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

activity_id required	integer (Activity Id)

Responses

— 204

Successful Response

> 422

Validation Error

DELETE /activities/{activity_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Daily Metrics

Create Daily Metric

Create a daily metric record.

If a record for the same (pid, date) already exists, returns 409 Conflict and advises the client to update the existing record via PUT /daily-metrics/{id} instead.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

pid required	integer (Pid)
date required	string <date> (Date)
calories_in >	Calories In (integer) or Calories In (null) (Calories In)
protein_g >	Protein G (integer) or Protein G (null) (Protein G)
sleep_hours >	Sleep Hours (number) or Sleep Hours (null) (Sleep Hours)

fatigue_level >	Fatigue Level (integer) or Fatigue Level (null) (Fatigue Level) Subjective fatigue 1-10
recovery >	Recovery (number) or Recovery (null) (Recovery) Recovery percentage
deep_work_hours >	Deep Work Hours (number) or Deep Work Hours (null) (Deep Work Hours)
stress_level >	Stress Level (integer) or Stress Level (null) (Stress Level) Subjective stress 1-10

Responses

> **201**

Successful Response

> **422**

Validation Error

POST /daily-metrics/

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": "2019-08-24",
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
```

```
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": "2019-08-24",
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1,
    "id": 0
}
```

List Daily Metrics

Retrieve daily metric records with optional filters.

`date_from` and `date_to` allow the front-end to request a specific window for charting (e.g. the last 30 days).

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`date_from` > Date From (string) or Date From (null) (Date From)
Start of date range (inclusive)

date_to >	Date To (string) or Date To (null) (Date To) End of date range (inclusive)
skip	integer (Skip) >= 0 Default: 0
limit	integer (Limit) [1 .. 200] Default: 50

Responses

> 200

Successful Response

> 422

Validation Error

GET /daily-metrics/

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
[
  - {
        "pid": 0,
        "date": "2019-08-24",
        "calories_in": 0,
        "protein_g": 0,
        "sleep_hours": 0,
        "fatigue_level": 1,
        "recovery": 100,
        "deep_work_hours": 0,
        "stress_level": 1,
        "id": 0
    }
]
```

Get Daily Metric

Retrieve a single daily metric record by primary key.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`metric_id` required	integer (Metric Id)

Responses

> **200**

Successful Response

> **422**

Validation Error

GET /daily-metrics/{metric_id}

Response samples

200 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "date": "2019-08-24",
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1,
    "id": 0
}
```

Update Daily Metric

Update an existing daily metric. Only fields present in the request body are modified (partial update via `exclude_unset`).

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

metric_id required	integer (Metric Id)

REQUEST BODY SCHEMA: application/json
required

pid >	Pid (integer) or Pid (null) (Pid)
date	null (Date)
calories_in >	Calories In (integer) or Calories In (null) (Calories In)
protein_g >	Protein G (integer) or Protein G (null) (Protein G)
sleep_hours >	Sleep Hours (number) or Sleep Hours (null) (Sleep Hours)
fatigue_level >	Fatigue Level (integer) or Fatigue Level (null) (Fatigue Level)
recovery >	Recovery (number) or Recovery (null) (Recovery)
deep_work_hours >	Deep Work Hours (number) or Deep Work Hours (null) (Deep Work Hours)
stress_level >	Stress Level (integer) or Stress Level (null) (Stress Level)

Responses

> **200**

Successful Response

> **422**

Validation Error

PUT /daily-metrics/{metric_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": null,
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1
}
```


Response samples

200 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "date": "2019-08-24",
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1,
    "id": 0
}
```

Delete Daily Metric

Delete a daily metric record. Returns 204 No Content on success.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

metric_id required	integer (Metric Id)

Responses

— 204

Successful Response

> 422

Validation Error

DELETE /daily-metrics/{metric_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Update Daily Metric By Date

Update a daily metric by date composite key.

This is often more convenient than updating by ID because the client naturally knows the user and date but may not know the row ID. Only fields present in the request body are modified (partial update).

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`date` required	string <date> (Date) Date of the metric (YYYY-MM-DD)

REQUEST BODY SCHEMA: application/json
required

`pid` >	Pid (integer) or Pid (null) (Pid)
`date`	null (Date)
`calories_in` >	Calories In (integer) or Calories In (null) (Calories In)
`protein_g` >	Protein G (integer) or Protein G (null) (Protein G)
`sleep_hours` >	Sleep Hours (number) or Sleep Hours (null) (Sleep Hours)
`fatigue_level` >	Fatigue Level (integer) or Fatigue Level (null) (Fatigue Level)
`recovery` >	Recovery (number) or Recovery (null) (Recovery)
`deep_work_hours` >	Deep Work Hours (number) or Deep Work Hours (null) (Deep Work Hours)
`stress_level` >	Stress Level (integer) or Stress Level (null) (Stress Level)

Responses

> **200**

Successful Response

> **422**

Validation Error

PUT /daily-metrics/by-date

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": null,
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": "2019-08-24",
    "calories_in": 0,
    "protein_g": 0,
    "sleep_hours": 0,
    "fatigue_level": 1,
    "recovery": 100,
    "deep_work_hours": 0,
    "stress_level": 1,
```

```
    "id": 0
}
```

Physiology Logs

Create Physiology Log

Create a new physiology snapshot for a user on a given date.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

pid required	integer (Pid)
date required	string <date> (Date)
height_cm >	Height Cm (number) or Height Cm (null) (Height Cm)
weight_kg >	Weight Kg (number) or Weight Kg (null) (Weight Kg)
running_fitness >	Running Fitness (number) or Running Fitness (null) (Running Fitness)
vo2max >	Vo2Max (number) or Vo2Max (null) (Vo2Max)
resting_heart_rate >	Resting Heart Rate (integer) or Resting Heart Rate (null) (Resting Heart Rate)
threshold_hr_zones >	Threshold Hr Zones (string) or Threshold Hr Zones (null) (Threshold Hr Zones) JSON string, e.g. {"z1":"120-135"}
threshold_pace_zones >	Threshold Pace Zones (string) or Threshold Pace Zones (null) (Threshold Pace Zones)

Responses

> 201

Successful Response

> 422

Validation Error

POST /physiology/

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": "2019-08-24",
    "height_cm": 0,
    "weight_kg": 0,
    "running_fitness": 0,
    "vo2max": 0,
    "resting_heart_rate": 0,
    "threshold_hr_zones": "string",
    "threshold_pace_zones": "string"
}
```

Response samples

201 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "date": "2019-08-24",
    "height_cm": 0,
    "weight_kg": 0,
    "running_fitness": 0,
    "vo2max": 0,
    "resting_heart_rate": 0,
    "threshold_hr_zones": "string",
    "threshold_pace_zones": "string",
    "id": 0
}
```

List Physiology Logs

Retrieve physiology logs scaled to the currently authenticated user.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`skip`	integer (Skip) `>= 0` Default: `0`
`limit`	integer (Limit) `[ 1 .. 200 ]` Default: `50`

Responses

> **200**

Successful Response

> 422

Validation Error

GET /physiology/

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
  - {
      "pid": 0,
      "date": "2019-08-24",
      "height_cm": 0,
      "weight_kg": 0,
      "running_fitness": 0,
      "vo2max": 0,
      "resting_heart_rate": 0,
      "threshold_hr_zones": "string",
      "threshold_pace_zones": "string",
      "id": 0
    }
]
```

Get Physiology Log

Retrieve a single physiology log by primary key, scoped to current user.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

log_id required	integer (Log Id)

Responses

200

Successful Response

422

Validation Error

GET /physiology/{log_id}

Response samples

200 422

Content type
application/json

Copy


```
{
    "pid": 0,
    "date": "2019-08-24",
    "height_cm": 0,
    "weight_kg": 0,
    "running_fitness": 0,
    "vo2max": 0,
    "resting_heart_rate": 0,
    "threshold_hr_zones": "string",
    "threshold_pace_zones": "string",
    "id": 0
}
```

Update Physiology Log

Partially update an existing physiology log.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`log_id` required	integer (Log Id)

REQUEST BODY SCHEMA: application/json
required

`pid` >	Pid (integer) or Pid (null) (Pid)
`date`	null (Date)
`height_cm` >	Height Cm (number) or Height Cm (null) (Height Cm)
`weight_kg` >	Weight Kg (number) or Weight Kg (null) (Weight Kg)
`running_fitness` >	Running Fitness (number) or Running Fitness (null) (Running Fitness)
`vo2max` >	Vo2Max (number) or Vo2Max (null) (Vo2Max)
`resting_heart_rate` >	Resting Heart Rate (integer) or Resting Heart Rate (null) (Resting Heart Rate)
`threshold_hr_zones` >	Threshold Hr Zones (string) or Threshold Hr Zones (null) (Threshold Hr Zones)
`threshold_pace_zones` >	Threshold Pace Zones (string) or Threshold Pace Zones (null) (Threshold Pace Zones)

Responses

> **200**

Successful Response

> 422

Validation Error

PUT /physiology/{log_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": null,
    "height_cm": 0,
    "weight_kg": 0,
    "running_fitness": 0,
    "vo2max": 0,
    "resting_heart_rate": 0,
    "threshold_hr_zones": "string",
    "threshold_pace_zones": "string"
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
{
    "pid": 0,
    "date": "2019-08-24",
    "height_cm": 0,
    "weight_kg": 0,
```

```
    "running_fitness": 0,
    "vo2max": 0,
    "resting_heart_rate": 0,
    "threshold_hr_zones": "string",
    "threshold_pace_zones": "string",
    "id": 0
}
```

Delete Physiology Log

Delete a physiology log. Returns 204 No Content on success.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`log_id` required	integer (Log Id)

Responses

— 204

Successful Response

> 422

Validation Error

DELETE `/physiology/{log_id}`

Response samples

422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "detail": [
      + { … }
    ]
}
```

Analytics

Physiology trend data for line charts + race predictions

Returns physiological trend data (VO2Max, resting HR, running fitness, weight) ordered chronologically for direct consumption by front-end charting libraries.

Also returns the latest threshold zones and race-time predictions based on the most recent VO2Max value.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`limit` integer (Limit) [1 .. 100]
Default: `12`
Number of recent snapshots

Responses

> **200**

Successful Response

> 422

Validation Error

GET /analytics/physiology/trends

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "trends": [
      + { … }
    ],
  - "current_status": {
        "date": "2019-08-24",
        "threshold_hr_zones": { },
        "threshold_pace_zones": { }
    },
  - "race_predictor": {
        "vo2max_used": 0,
        "predicted_5k": "string",
        "predicted_10k": "string",
        "predicted_half_marathon": "string"
    }
}
```


Personal records (PRs) — best times for standard race distances

Returns the fastest finish time for each standard race distance:

Running: 5K, 10K, Half Marathon **Cycling:** 10km, 50km, 100km

Activities are matched to target distances using tolerance bands (e.g. 5K = 4.8–5.5 km). Anomalies are filtered out:

- Run pace < 2:30/km or > 10:00/km → excluded
- Ride average speed > 70 km/h or < 8 km/h → excluded

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200**

Successful Response

GET /analytics/performance/records

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
{
  - "run_records": [
      + { … }
    ],
  - "ride_records": [
      + { … }
    ]
}
```

Daily training load calendar + intensity distribution pie chart

Aggregates Activity data over the past N days:

- Per-day summary of training load, run km, ride km (for calendar / bar chart)
- Period totals
- Intensity distribution by average heart rate (Easy / Tempo / Hard) for pie chart rendering

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`days` integer (Days) [1 .. 365]
Default: 30
Lookback window in days

Responses

> **200**

Successful Response

> **422**

Validation Error

GET /analytics/training/status

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "daily_summary": [
      + { … }
    ],
  - "totals": {
        "total_load": 0,
        "total_run_km": 0,
        "total_ride_km": 0,
        "total_activities": 0,
        "total_duration_min": 0
    },
  - "intensity_distribution": [
      + { … }
    ]
}
```

How temperature affects running performance

Segments running activities by temperature into Cold (< 10°C), Moderate (10–22°C), and Hot (> 22°C), then computes average heart rate and pace for each bucket.

Only includes records with both temperature and avg_heart_rate populated to avoid NULL contamination in averages.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200**

Successful Response

GET /analytics/insights/environment

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
{
  - "temperature_zones": [
      + { … }
    ]
}
```

How sleep and fatigue correlate with running performance

Joins Activity (Run) with same-day DailyMetric to analyse:

1. Sleep impact: ≥ 7.5h sleep vs < 7.5h → difference in pace and HR
2. Fatigue impact: fatigue ≥ 7 vs < 7 → difference in HR and load

Only activities with a matching DailyMetric on the same (pid, date) are included. NULL values in the compared fields are filtered out.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> 200

Successful Response

GET /analytics/insights/lifestyle

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
{
  - "sleep_impact": [
      + { … }
    ],
  - "fatigue_impact": [
      + { … }
    ]
}
```

Health

Health Check

Simple health-check endpoint to verify the API is running.

Responses

> **200**

Successful Response

GET /

Response samples

200

Content type

application/json

Copy

```
null
```